AE 462
Lab 7
Lab Session: Wednesday, April 1, 2026
Due: Wednesday, April 8, 2026, 9am CST

This laboratory assignment will focus on three-dimensional flow simulation and postprocessing. The items required for your assignment submission are given here, alongside helpful instructions for completing this lab. As with previous assignments, all plots and presentations of data should include units, axis titles, and legends where appropriate. Provide your responses in a single PDF file. Upload any scripts, spreadsheets, or other work used during data analysis.

This laboratory exercise considers the ONERA M6 swept wing in transonic flow. This configuration features the following geometric properties:

- Semispan, b = 1.1963 m
- Mean Aerodynamic Chord, $\bar{c}$ = 0.64607 m
- Aspect Ratio, AR = 3.8
- Taper Ratio, λ = 0.562
- Leading-edge sweep, Λ_{LE} = 30.0 deg
- Trailing-edge sweep, Λ_{TE} = 15.8 deg

Additionally, we will be simulating this configuration with the following freestream conditions:

- Freestream Mach number, M_∞ = 0.84
- Angle of attack, α = 3.06 deg
- Angle of sideslip, β = 0 deg
- Inviscid flow

1) **Aerodynamic Performance**
   a. Report your converged values of lift coefficient and drag coefficient for this configuration.
   b. ONERA never published the experimental lift and drag coefficients for the wind tunnel campaign for this wing. However, it is a common configuration for benchmarking CFD workflows. Compare your CFD predictions of C_L and C_D to those shown in the table below. If differences are observed, comment on any anticipated reasons for the discrepancies observed.

Solver	Turb. Model	Flux Scheme	Precision	Mesh Type	C_l	C_d
DG-DES	S-A	Roe	Double	Unstructured	0.2579	0.01970
DG-DES	DES	Roe	Double	Unstructured	0.2564	0.01960
DG-DES	DDES	Roe	Double	Unstructured	0.2580	0.01955
DG-DES	DES	AUSM	Double	Unstructured	0.2540	0.01910
DG-DES	DES	HLLC	Double	Unstructured	0.2550	0.01991
MERLIN[22]	S-A	Osher	Double	Structured	0.2697	0.01736
Neilsen et al.[24]	S-A	xx	xx	Unstructured	0.2530	0.01680
Lee et. al.[23]	SST	xx	xx	Structured	0.2622	0.01751

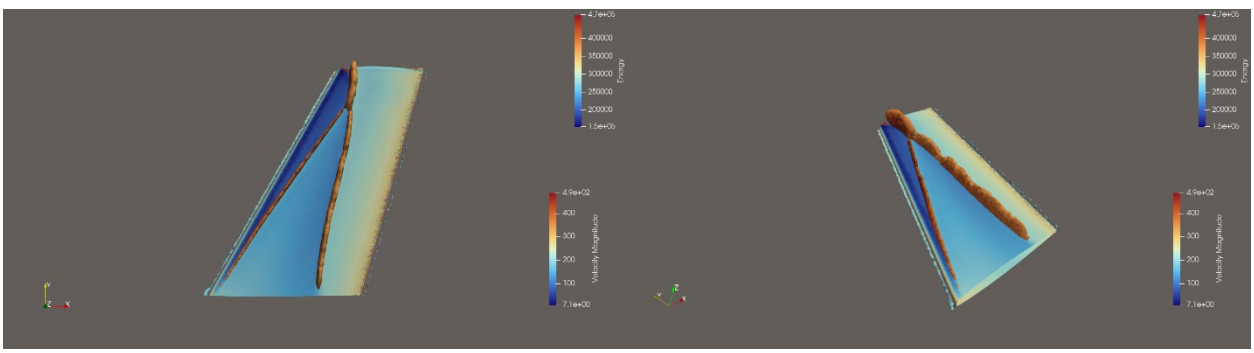
2) **Comparisons with Experimental Pressure Distribution**

a. Compare the pressure coefficient distributions produced by your inviscid simulation of the ONERA M6 wing to those obtained from wind tunnel experiments. Co-plot your C_p distributions with the experimental values for spanwise positions of y/b = 0.2, 0.44, 0.65, 0.8, 0.9, 0.95, and 0.99. Provide separate plots for each spanwise plane considered (7 plots total). Use the local chord length produced at the given spanwise location when calculating the x/c values. Plot your C_p distributions using only symbols (without lines).

b. For these seven spanwise positions, extract a slice of the Mach number field produced across the streamwise flow plane in Paraview and provide a snapshot alongside these C_p distributions.

3) **Shock Flow Visualization**

a. From Paraview, provide a visualization of the transonic shocks across the upper surface of the wing using the gradient of the density field with respect to x (threshold of +5 kg/m^4). Color the resulting isosurfaces by Mach number. Also, include the surface pressure coefficient distribution produced across the 3D wing to observe how the shock structure strongly influences the wing surface pressure. Provide a top-down and isometric view, which should result in figures that resemble the two images shown below. Note that these figures provided use slightly different parameters, but they will look visually consistent with your result.



4) **Wingtip Flow**

a. To visualize the flow produced by the tip vortex, add a stream tracer filter in the flow, starting and ending at x, y, and z positions of [0, 1.1, -0.06] and [0, 1.2, -0.06], respectively. Color the stream tracer by Rotation. Include the surface Mach number distribution produced by the wing. Provide two snapshots of the resulting flow field, similar to the examples shown for 3) above.

Lab Instructions:

Organizing files

Create a new folder within your U:\AE 462 directory called "Lab 7". Download the files "3D_mesh.py", "AileM6_with_sharp_TE.stp", "Onera.cfg", and "OneraM6_Exp.zip". Place these files in your U:\AE 462\Lab 7 directory.

Creating a Mesh

For this assignment, you will be using a Python script to generate your mesh. The 3D_mesh.py script is already configured to call in the ONERA M6 STEP file and mesh the configuration in 3D. Run this Python script. After some time, a Gmsh window will be generated, and you will see your mesh in the graphical interface. Export your mesh as usual in .su2 format. The Onera.cfg file is configured to read in a mesh file called "M6.su2", but feel free to use whatever naming convention you prefer.

Running the Solver

Run SU2 in parallel to solve the inviscid flow around the ONERA M6 wing. The default configuration file is set up to use multi-grid cycles to accelerate convergence, so your solution should take less than 150 iterations. Save the converged C_L and C_D values for future reporting, or extract them from the history.csv file.

Post-Processing Field Data

Load the flow.vtu file *and* the surface_flow.vtu files into Paraview. You will apply filters to both the field data and surface data to complete this assignment.

To extract surface pressure distributions, apply the "Slice" filter on the surface_flow.vtu item in the Pipeline Browser. To obtain data at fixed spanwise (y) planes, under Plane Parameters set the Origin to [0, Y, 0] where Y is the spanwise position (in m) where you would like to extract data. You will need to obtain surface data across planes y = 0.23926 m, 0.526372 m, 0.777595 m, 0.95704 m, 1.07667 m, 1.136485 m, and 1.184337 m. Also under Plane Parameters, set Normal to [0, 1, 0]. Click "Apply". You can now save the surface data at this plane using the "Save Data" function in the toolbar with the Slice filter selected in the Pipeline Browser, or plot within Paraview using the "Plot Data" filter. You will need these slice extractions of surface data to co-plot the C_p distributions produced by the wing to those obtained from experiments.

For this lab exercise, you will also be comparing the simulated surface C_p distributions to those obtained from experiments. The experimental surface pressures are provided in files that follow a naming convention of, "cp#u.ex" and "cp#l.ex" where # is the spanwise row (e.g., 1 corresponds to y = 0.23926 m) and u or l indicate the upper or lower surface. In these files, the first column represents the x/c position and the second column represents the $-C_p$ value measured at this position.

You can also use the Slice filter to the off-body flow field data. Apply the "Slice" filter on the flow.vtu item in the Pipeline Browser. Apply the same Origin and Normal strategy used for the surface field data earlier. Apply the Slice filter initially to the y = 0.23926 m plane. Color the surface by Mach number. You can orient the camera in Paraview to be normal to the y axis by

clicking the +Y view direction

. Save a snapshot of your Mach number field, and repeat this process to obtain Mach number fields for the other six spanwise planes considered earlier.

Now you will co-plot features of the surface and volume flow fields together. Start by ensuring that both the flow.vtu and surface_flow.vtu items are visible in the Pipeline Browser. Color the surface_flow.vtu by Pressure Coefficient. Now, we will filter the flow.vtu off-body flow field to extract the transonic shock structures produced in the flow simulation. Go to the flow.vtu item in the Pipeline Browser and select the "Gradient" filter. Under the Properties window of this filter, change the Scalar Array to Density. Click "Apply". With the Gradient filter selected, which you have just created, apply a "Contour" filter. Under the "Contour by" option in this filter, select Gradient_X to represent the *x*-component of the gradient in the density field. Under "Isosurfaces", change the default value to 5 and click "Apply". You should now see the transonic shocks appear over the wing surface. Change the coloring of these isocontours to Mach. Take two snapshots of the wing, one looking just at the wing upper surface and one that provides an isometric view of the wing (see the example in the problem statement for 3) above). You can orient the camera normal to the wing upper surface by selecting the -Z option in the toolbar above

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Change the surface_flow.vtu representation to Mach. Disable the density gradient contour so it is not visible in the flow field. Return to the flow.vtu item in the Pipeline Browser. To this set of flow data, apply a "Stream Tracer" filter. Under the Line Parameters option, change the Point 1 and Point 2 values to: [0, 1.1, -0.06] and [0, 1.2, -0.06], respectively. Click "Apply". Change the coloring of these stream traces to Rotation. Take two snapshots of the wing flow field similar to that done previously.

You now have the CFD-generated data needed to respond to the items required for the submission.